



Hackathon Project Submission – Database Administration

Student Database Management System

Student Details

Student Name	Ashifa
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Department	BCA
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College	Government Arts and Science College, Gudalur
Date of Submission	23-09-2026
GitHub Link	

1. Problem Statement & Objectives

Problem Statement

In many educational institutions, student information such as personal details, department details, course enrollment, examination schedules, and academic marks is maintained manually or across separate systems. This can lead to difficulties in storing, updating, retrieving, and managing student records efficiently.

The **Student Database Management System** is developed to provide a centralized database for managing student-related information. The system stores student details, departments, courses, enrollments, marks, faculty information, and examination details in a structured relational database.

The system uses **MySQL** to maintain accurate and organized records and allows users to perform operations such as inserting, updating, deleting, searching, sorting, grouping, and



retrieving related information using SQL queries. The project aims to reduce manual data management and make student information easier to access and manage.

Objectives

The main objectives of the **Student Database Management System** are:

1. **To store student information** such as name, date of birth, gender, phone number, email, and department.
2. **To manage department details** including department names and Head of Department information.
3. **To maintain course information** such as course name, course code, credits, and department.
4. **To manage student enrollment** in different courses and semesters.
5. **To record and manage academic marks** including internal, external, and total marks.
6. **To maintain faculty information** such as faculty name, designation, contact details, and department.
7. **To manage examination details** including exam name, type, date, and associated course.
8. **To retrieve information efficiently** using SQL queries such as WHERE, ORDER BY, LIKE, BETWEEN, GROUP BY, aggregate functions, and INNER JOIN.
9. **To reduce manual data management** by maintaining student records in a structured relational database.
10. **To ensure data consistency and integrity** using primary keys, foreign keys, constraints, and appropriate data types.



2. Database Design

2.1 Database Structure

The **Student Database Management System** is designed using **MySQL**. The database is named `Student_Database_Management` and consists of **7 tables** that store and manage different types of student and academic information.

Tables Used

The Student Database Management System uses **7 tables**:

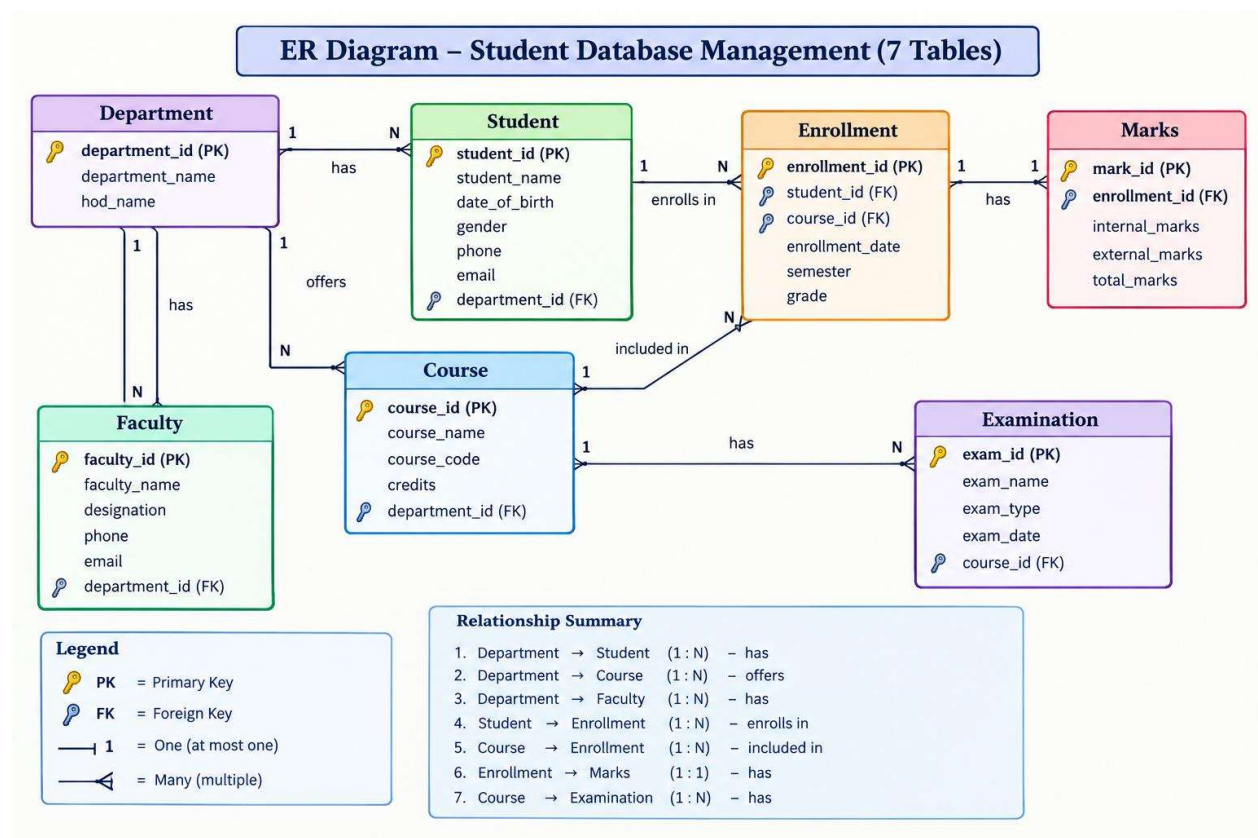
1. **Department** – Stores department ID, department name, and HOD name.
2. **Student** – Stores student personal, contact, and department details.
3. **Course** – Stores course name, course code, credits, and department.
4. **Enrollment** – Stores student enrollment, course, semester, and grade details.
5. **Marks** – Stores internal, external, and total marks of students.
6. **Faculty** – Stores faculty details, designation, contact information, and department.
7. **Examination** – Stores examination name, type, date, and associated course.

Table	Primary Key	Main Purpose
Department	department_id	Department information
Student	student_id	Student information
Course	course_id	Course information
Enrollment	enrollment_id	Student-course enrollment
Marks	mark_id	Student marks
Faculty	faculty_id	Faculty information
Examination	exam_id	Examination information

2.2 Entity Relationship Diagram

The ER diagram represents the relationships between the seven tables in the Student Database Management System. It shows the primary keys, foreign keys, and how student, department, course, enrollment, marks, faculty, and examination data are connected.

Diagram:





2.3 Technologies Used

- ☐ **MySQL** – Database management and SQL operations
- ☐ **MySQL Workbench** – Database design, table creation, and query execution
- ☐ **SQL** – Creating tables, inserting data, updating records, and retrieving information
- ☐ **Microsoft Word** – Preparing the project report
- ☐ **GitHub** – Storing and sharing the project files and source code

3. Practical Implementation

3.1 Database and Table Creation

Database Creation

```
CREATE DATABASE Student_Database_Management;
```

```
USE Student_Database_Management;
```

Department Table

```
CREATE TABLE Department (  
    department_id INT PRIMARY KEY AUTO_INCREMENT,  
    department_name VARCHAR(100) NOT NULL,  
    hod_name VARCHAR(100)  
);
```



Student Table

```
CREATE TABLE Student (  
    student_id INT PRIMARY KEY AUTO_INCREMENT,  
    student_name VARCHAR(100) NOT NULL,  
    date_of_birth DATE,  
    gender VARCHAR(10),  
    phone VARCHAR(15),  
    email VARCHAR(100),  
    department_id INT NOT NULL,  
    FOREIGN KEY (department_id) REFERENCES Department(department_id)  
);
```

Course Table

```
CREATE TABLE Course (  
    course_id INT PRIMARY KEY AUTO_INCREMENT,  
    course_name VARCHAR(100) NOT NULL,  
    course_code VARCHAR(20) NOT NULL UNIQUE,  
    credits INT NOT NULL,  
    department_id INT NOT NULL,  
    FOREIGN KEY (department_id) REFERENCES Department(department_id)  
);
```

Enrollment Table

```
CREATE TABLE Enrollment (  

```



```
enrollment_id INT PRIMARY KEY AUTO_INCREMENT,  
student_id INT NOT NULL,  
course_id INT NOT NULL,  
enrollment_date DATE,  
semester INT,  
grade VARCHAR(5),  
FOREIGN KEY (student_id) REFERENCES Student(student_id),  
FOREIGN KEY (course_id) REFERENCES Course(course_id)  
);
```

Marks Table

```
CREATE TABLE Marks (  
mark_id INT PRIMARY KEY AUTO_INCREMENT,  
enrollment_id INT NOT NULL,  
internal_marks DECIMAL(5,2),  
external_marks DECIMAL(5,2),  
total_marks DECIMAL(5,2),  
FOREIGN KEY (enrollment_id) REFERENCES Enrollment(enrollment_id)  
);
```

Faculty Table

```
CREATE TABLE Faculty (
```



```
faculty_id INT PRIMARY KEY AUTO_INCREMENT,  
faculty_name VARCHAR(100) NOT NULL,  
designation VARCHAR(50),  
phone VARCHAR(15),  
email VARCHAR(100),  
department_id INT NOT NULL,  
FOREIGN KEY (department_id) REFERENCES Department(department_id)  
);
```

Examination Table

```
CREATE TABLE Examination (  
    exam_id INT PRIMARY KEY AUTO_INCREMENT,  
    exam_name VARCHAR(100) NOT NULL,  
    exam_type VARCHAR(50),  
    exam_date DATE,  
    course_id INT NOT NULL,  
    FOREIGN KEY (course_id) REFERENCES Course(course_id)  
);
```

3.2 Constraints and Relationships

Constraints Used

- **PRIMARY KEY** – Uniquely identifies each record in a table.
- **FOREIGN KEY** – Establishes relationships between tables and maintains referential integrity.



- **NOT NULL** – Ensures that important fields cannot be left empty.
- **UNIQUE** – Ensures that values such as course codes are not duplicated.
- **AUTO_INCREMENT** – Automatically generates unique ID values.

Relationships

Relationship	Type
Department → Student	One-to-Many
Department → Course	One-to-Many
Department → Faculty	One-to-Many
Student → Enrollment	One-to-Many
Course → Enrollment	One-to-Many
Enrollment → Marks	One-to-Many*
Course → Examination	One-to-Many

3.3 Data Insertion

Department Data

INSERT INTO Department (department_name, hod_name)

VALUES

('Computer Applications', 'Dr. Ramesh Kumar'),

('Computer Science', 'Dr. Priya Sharma'),

('Information Technology', 'Dr. Arun Kumar'),

('Commerce', 'Dr. Meena Devi'),

('Mathematics', 'Dr. Suresh Babu');

Student Data

INSERT INTO Student

(student_name, date_of_birth, gender, phone, email, department_id)



VALUES

('Ananya Raj', '2004-05-12', 'Female', '9876543210', 'ananya@gmail.com', 1),
('Arjun Kumar', '2003-08-21', 'Male', '9876543211', 'arjun@gmail.com', 1),
('Meera Nair', '2004-01-15', 'Female', '9876543212', 'meera@gmail.com', 2),
('Rahul Das', '2003-11-30', 'Male', '9876543213', 'rahul@gmail.com', 2),
('Sneha Thomas', '2004-07-09', 'Female', '9876543214', 'sneha@gmail.com', 3),
('Vishnu Menon', '2003-03-18', 'Male', '9876543215', 'vishnu@gmail.com', 3),
('Keerthana S', '2004-09-25', 'Female', '9876543216', 'keerthana@gmail.com', 1),
('Adithya P', '2003-12-05', 'Male', '9876543217', 'adithya@gmail.com', 4),
('Fathima Ali', '2004-06-14', 'Female', '9876543218', 'fathima@gmail.com', 1),
('Naveen Kumar', '2003-10-11', 'Male', '9876543219', 'naveen@gmail.com', 5);

Course Data

INSERT INTO Course

(course_name, course_code, credits, department_id)

VALUES

('Database Management Systems', 'BCA301', 4, 1),
('Web Technology', 'BCA302', 4, 1),
('Python Programming', 'BCA303', 4, 1),
('Computer Networks', 'BCA304', 3, 1),
('Operating Systems', 'BCA305', 4, 1),
('Data Structures', 'CSC301', 4, 2),
('Java Programming', 'CSC302', 4, 2),
('Information Security', 'IT301', 3, 3),



('Financial Accounting', 'COM301', 4, 4),

('Discrete Mathematics', 'MAT301', 4, 5);

Enrollment Data

INSERT INTO Enrollment

(student_id, course_id, enrollment_date, semester, grade)

VALUES

(1, 1, '2026-06-10', 5, 'A'),

(1, 2, '2026-06-10', 5, 'A'),

(1, 3, '2026-06-10', 5, 'B+'),

(2, 1, '2026-06-11', 5, 'B+'),

(2, 4, '2026-06-11', 5, 'A'),

(3, 6, '2026-06-12', 5, 'A'),

(3, 7, '2026-06-12', 5, 'A'),

(4, 6, '2026-06-12', 5, 'B'),

(4, 7, '2026-06-12', 5, 'B+'),

(5, 8, '2026-06-13', 5, 'A'),

(6, 8, '2026-06-13', 5, 'B+'),

(7, 1, '2026-06-14', 5, 'A'),

(7, 5, '2026-06-14', 5, 'A'),

(8, 9, '2026-06-15', 5, 'B+'),

(9, 1, '2026-06-15', 5, 'A+'),

(9, 3, '2026-06-15', 5, 'A'),

(10, 10, '2026-06-16', 5, 'A');



Marks Data

INSERT INTO Marks

(enrollment_id, internal_marks, external_marks, total_marks)

VALUES

(1, 24, 68, 92),

(2, 22, 64, 86),

(3, 20, 61, 81),

(4, 21, 60, 81),

(5, 23, 67, 90),

(6, 25, 70, 95),

(7, 24, 66, 90),

(8, 19, 58, 77),

(9, 22, 63, 85),

(10, 25, 69, 94),

(11, 21, 62, 83),

(12, 24, 71, 95),

(13, 23, 68, 91),

(14, 20, 59, 79),

(15, 25, 73, 98),

(16, 23, 67, 90),

(17, 24, 70, 94);

Faculty Data

INSERT INTO Faculty



(faculty_name, designation, phone, email, department_id)

VALUES

('Dr. Ramesh Kumar', 'Professor', '9876500010', 'ramesh@college.com', 1),
('Dr. Priya Sharma', 'Associate Professor', '9876500011', 'priya@college.com', 2),
('Dr. Arun Kumar', 'Assistant Professor', '9876500012', 'arun@college.com', 3),
('Dr. Meena Devi', 'Professor', '9876500013', 'meena@college.com', 4),
('Dr. Suresh Babu', 'Associate Professor', '9876500014', 'suresh@college.com', 5),
('Prof. Kavitha Raj', 'Assistant Professor', '9876500015', 'kavitha@college.com', 1),
('Prof. Manoj Das', 'Assistant Professor', '9876500016', 'manoj@college.com', 2);

Examination Data

INSERT INTO Examination

(exam_name, exam_type, exam_date, course_id)

VALUES

('Database Management Systems Exam', 'End Semester', '2026-11-10', 1),
('Web Technology Exam', 'End Semester', '2026-11-12', 2),
('Python Programming Exam', 'End Semester', '2026-11-15', 3),
('Computer Networks Exam', 'End Semester', '2026-11-18', 4),
('Operating Systems Exam', 'End Semester', '2026-11-20', 5),
('Data Structures Exam', 'End Semester', '2026-11-22', 6),
('Java Programming Exam', 'End Semester', '2026-11-25', 7),
('Information Security Exam', 'End Semester', '2026-11-27', 8),
('Financial Accounting Exam', 'End Semester', '2026-11-29', 9),
('Discrete Mathematics Exam', 'End Semester', '2026-12-02', 10);



3.4 SQL Operations

INSERT

INSERT INTO Student

(student_name, date_of_birth, gender, phone, email, department_id)

VALUES

('Rahul Menon', '2004-02-20', 'Male', '9876543220', 'rahul.menon@gmail.com', 1);

SELECT

SELECT * FROM Student;

UPDATE

UPDATE Student

SET phone = '9998887770'

WHERE student_id = 1;

DELETE

DELETE FROM Student

WHERE student_id = 11;

WHERE

SELECT *

FROM Student

WHERE department_id = 1;



ORDER BY

```
SELECT *  
FROM Student  
ORDER BY student_name DESC;
```

LIKE

```
SELECT *  
FROM Student  
WHERE student_name LIKE 'A%';
```

BETWEEN

```
SELECT *  
FROM Student  
WHERE date_of_birth BETWEEN '2004-01-01' AND '2004-12-31';
```

Aggregate Functions

- **COUNT()** - Counts the number of records

```
SELECT COUNT(*) AS total_records  
FROM Marks;
```

- **SUM()** - Calculates the total marks

```
SELECT SUM(total_marks) AS total_marks  
FROM Marks;
```

- **AVG()** - Calculates the average marks
SELECT AVG(total_marks) AS average_marks
FROM Marks;



- **MAX()** - Finds the highest marks
SELECT MAX(total_marks) AS highest_marks
FROM Marks;
- **MIN()** - Finds the lowest marks
SELECT MIN(total_marks) AS lowest_marks
FROM Marks;

GROUP BY

```
SELECT department_id, COUNT(*) AS total_students  
FROM Student  
GROUP BY department_id;
```

INNER JOIN

```
SELECT  
    Student.student_name,  
    Course.course_name,  
    Enrollment.semester,  
    Enrollment.grade  
FROM Student  
INNER JOIN Enrollment  
    ON Student.student_id = Enrollment.student_id  
INNER JOIN Course  
    ON Enrollment.course_id = Course.course_id;
```

Alter Table

ALTER TABLE Student

ADD COLUMN address VARCHAR(150);

3.5 Project Execution

The Student Database Management System was executed using MySQL Workbench. The database was created first, followed by the creation of seven related tables: Department, Student, Course, Enrollment, Marks, Faculty, and Examination.

Sample records were then inserted into each table. Various SQL operations were executed to test the database, including SELECT, INSERT, UPDATE, DELETE, WHERE, ORDER BY, LIKE, BETWEEN, GROUP BY, aggregate functions, INNER JOIN, and ALTER TABLE. The execution results were verified in MySQL Workbench to ensure that the database functions correctly and the required student information can be stored, updated, and retrieved efficiently.

4. Output Screenshots

Screenshot 1 – Database and Tables

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▶ sys	

Screenshot 2 – Department Table with Records

232 ON Enrollment.course_id = Course.course_id;
233 • SELECT * FROM Department;

Result Grid Filter Rows: Edit:

department_id	department_name	hod_name
1	Computer Applications	Dr. Ramesh Kumar
2	Computer Science	Dr. Priya Sharma
3	Information Technology	Dr. Arun Kumar
4	Commerce	Dr. Meena Devi
5	Mathematics	Dr. Suresh Babu
NULL	NULL	NULL

Department 21 x

Screenshot 3 – Student Table with Records

233 • `SELECT * FROM Department;`
234 • `SELECT * FROM Student;`

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

student_id	student_name	date_of_birth	gender	phone	email	department_id
1	Ananya Raj	2004-05-12	Female	9998887770	ananya@gmail.com	1
2	Arjun Kumar	2003-08-21	Male	9876543211	arjun@gmail.com	1
3	Meera Nair	2004-01-15	Female	9876543212	meera@gmail.com	2
4	Rahul Das	2003-11-30	Male	9876543213	rahul@gmail.com	2
5	Sneha Thomas	2004-07-09	Female	9876543214	sneha@gmail.com	3
6	Vishnu Menon	2003-03-18	Male	9876543215	vishnu@gmail.com	3
7	Keerthana S	2004-09-25	Female	9876543216	keerthana@gmail.com	1

Student 22 x

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Cont

student_id	student_name	date_of_birth	gender	phone	email	department_id
7	Keerthana S	2004-09-25	Female	9876543216	keerthana@gmail.com	1
8	Adithya P	2003-12-05	Male	9876543217	adithya@gmail.com	4
9	Fathima Ali	2004-06-14	Female	9876543218	fathima@gmail.com	1
10	Naveen Kumar	2003-10-11	Male	9876543219	naveen@gmail.com	5

Screenshot 4 – Course Table and Other Tables with Records

Course Table with Records

234 • `SELECT * FROM Student;`
235 • `SELECT * FROM Course;`

Result Grid | Filter Rows: | Edit: | Export/Import: |

course_id	course_name	course_code	credits	department_id
1	Database Management Systems	BCA301	4	1
2	Web Technology	BCA302	4	1
3	Python Programming	BCA303	4	1
4	Computer Networks	BCA304	3	1
5	Operating Systems	BCA305	4	1
6	Data Structures	CSC301	4	2
7	Java Programming	CSC302	4	2

Course 23 x

Output

234 • `SELECT * FROM Student;`
235 • `SELECT * FROM Course;`

Result Grid | Filter Rows: | Edit: | Export/Import:

	course_id	course_name	course_code	credits	department_id
	6	Data Structures	CSC301	4	2
	7	Java Programming	CSC302	4	2
	8	Information Security	IT301	3	3
	9	Financial Accounting	COM301	4	4
	10	Discrete Mathematics	MAT301	4	5
*	NULL	NULL	NULL	NULL	NULL

Course 23 x

Output

Enrollment Table with Records

235 • `SELECT * FROM Course;`
236 • `SELECT * FROM Enrollment;`

Result Grid | Filter Rows: | Edit: | Export/Import:

	enrollment_id	student_id	course_id	enrollment_date	semester	grade
▶	1	1	1	2026-06-10	5	A
	2	1	2	2026-06-10	5	A
	3	1	3	2026-06-10	5	B+
	4	2	1	2026-06-11	5	B+
	5	2	4	2026-06-11	5	A
	6	3	6	2026-06-12	5	A
	7	3	7	2026-06-12	5	A

Enrollment 24 x

Output

Marks Table with Records

236 • `SELECT * FROM Enrollment;`
237 • `SELECT * FROM Marks;`

Result Grid | Filter Rows: | Edit: | Export/

	mark_id	enrollment_id	internal_marks	external_marks	total_marks
▶	1	1	24.00	68.00	92.00
	2	2	22.00	64.00	86.00
	3	3	20.00	61.00	81.00
	4	4	21.00	60.00	81.00
	5	5	23.00	67.00	90.00
	6	6	25.00	70.00	95.00
	7	7	24.00	66.00	90.00

Marks 25 x

Output

Faculty Table with Records

237 • `SELECT * FROM Marks;`
238 • `SELECT * FROM Faculty;`

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

	faculty_id	faculty_name	designation	phone	email	department_id
▶	1	Dr. Ramesh Kumar	Professor	9876500010	ramesh@college.com	1
	2	Dr. Priya Sharma	Associate Professor	9876500011	priya@college.com	2
	3	Dr. Arun Kumar	Assistant Professor	9876500012	arun@college.com	3
	4	Dr. Meena Devi	Professor	9876500013	meena@college.com	4
	5	Dr. Suresh Babu	Associate Professor	9876500014	suresh@college.com	5
	6	Prof. Kavitha Raj	Assistant Professor	9876500015	kavitha@college.com	1
	7	Prof. Manoj Das	Assistant Professor	9876500016	manoj@college.com	2

Faculty 26 x

Output

Action Output

Time Action Message

Examination Table with Records

238 `SELECT * FROM Faculty;`
239 `SELECT * FROM Examination;`

Result Grid

	exam_id	exam_name	exam_type	exam_date	course_id
▶	1	Database Management Systems Exam	End Semester	2026-11-10	1
	2	Web Technology Exam	End Semester	2026-11-12	2
	3	Python Programming Exam	End Semester	2026-11-15	3
	4	Computer Networks Exam	End Semester	2026-11-18	4
	5	Operating Systems Exam	End Semester	2026-11-20	5
	6	Data Structures Exam	End Semester	2026-11-22	6
	7	Java Programming Exam	End Semester	2026-11-25	7

Examination 27 ✕

Screenshot 5 – INSERT / UPDATE / DELETE Operations

INSERT Operation

240 `INSERT INTO Student`
241 `(student_name, date_of_birth, gender, phone, email, department_id)`
242 `VALUES`
243 `('Rahul Menon', '2004-02-20', 'Male', '9876543220', 'rahul.menon@gmail.com', 1);`

Context Help Snippets

Output

#	Time	Action	Message	Duration / Fetch
35	17:01:10	SELECT * FROM Course LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
36	17:03:06	SELECT * FROM Enrollment LIMIT 0, 1000	17 row(s) returned	0.000 sec / 0.000 sec
37	17:05:22	SELECT * FROM Marks LIMIT 0, 1000	17 row(s) returned	0.000 sec / 0.000 sec
38	17:06:52	SELECT * FROM Faculty LIMIT 0, 1000	7 row(s) returned	0.016 sec / 0.000 sec
39	17:10:05	SELECT * FROM Examination LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
40	17:10:56	INSERT INTO Student (student_name, date_of_birth, gender, phone, email, department_id) ...	1 row(s) affected	0.015 sec

ENG 17:11

UPDATE Operation

```

244 • UPDATE Student
245 SET phone = '9998887770'
246 WHERE student_id = 1;
    
```

Output

#	Time	Action	Message	Duration / Fetch
36	17:03:06	SELECT * FROM Enrollment LIMIT 0, 1000	17 row(s) returned	0.000 sec / 0.000 sec
37	17:05:22	SELECT * FROM Marks LIMIT 0, 1000	17 row(s) returned	0.000 sec / 0.000 sec
38	17:06:52	SELECT * FROM Faculty LIMIT 0, 1000	7 row(s) returned	0.016 sec / 0.000 sec
39	17:10:05	SELECT * FROM Examination LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
40	17:10:56	INSERT INTO Student (student_name, date_of_birth, gender, phone, email, department_id) ...	1 row(s) affected	0.015 sec
41	17:11:53	UPDATE Student SET phone = '9998887770' WHERE student_id = 1	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.016 sec

DELETE Operation

```

247 • DELETE FROM Student
248 WHERE student_id = 11;
    
```

Output

#	Time	Action	Message	Duration / Fetch
37	17:05:22	SELECT * FROM Marks LIMIT 0, 1000	17 row(s) returned	0.000 sec / 0.000 sec
38	17:06:52	SELECT * FROM Faculty LIMIT 0, 1000	7 row(s) returned	0.016 sec / 0.000 sec
39	17:10:05	SELECT * FROM Examination LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
40	17:10:56	INSERT INTO Student (student_name, date_of_birth, gender, phone, email, department_id) ...	1 row(s) affected	0.015 sec
41	17:11:53	UPDATE Student SET phone = '9998887770' WHERE student_id = 1	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.016 sec
42	17:12:46	DELETE FROM Student WHERE student_id = 11	0 row(s) affected	0.000 sec

Screenshot 6 – SELECT and Filtering Operation

SELECT Operation

```

248 WHERE student_id = 11;
249 • SELECT * FROM Student;
    
```

Result Grid

student_id	student_name	date_of_birth	gender	phone	email	department_id
1	Ananya Raj	2004-05-12	Female	9998887770	ananya@gmail.com	1
2	Arjun Kumar	2003-08-21	Male	9876543211	arjun@gmail.com	1
3	Meera Nair	2004-01-15	Female	9876543212	meera@gmail.com	2
4	Rahul Das	2003-11-30	Male	9876543213	rahul@gmail.com	2
5	Sneha Thomas	2004-07-09	Female	9876543214	sneha@gmail.com	3
6	Vishnu Menon	2003-03-18	Male	9876543215	vishnu@gmail.com	3
7	Keerthana S	2004-09-25	Female	9876543216	keerthana@gmail.com	1

Student 28 x

Output

WHERE Operation

```

249 SELECT * FROM Student;
250 SELECT *
251 FROM Student
252 WHERE department_id = 1;
253
254

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: I

	student_id	student_name	date_of_birth	gender	phone	email	department_id
▶	1	Ananya Raj	2004-05-12	Female	9998887770	ananya@gmail.com	1
	2	Arjun Kumar	2003-08-21	Male	9876543211	arjun@gmail.com	1
	7	Keerthana S	2004-09-25	Female	9876543216	keerthana@gmail.com	1
	9	Fathima Ali	2004-06-14	Female	9876543218	fathima@gmail.com	1
	13	Rahul Menon	2004-02-20	Male	9876543220	rahul.menon@gmail.com	1
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Student 1 x

Output

ORDER BY Operation

```

253
254 SELECT *
255 FROM Student
256 ORDER BY student_name DESC;

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: I

	student_id	student_name	date_of_birth	gender	phone	email	department_id
▶	6	Vishnu Menon	2003-03-18	Male	9876543215	vishnu@gmail.com	3
	5	Sneha Thomas	2004-07-09	Female	9876543214	sneha@gmail.com	3
	13	Rahul Menon	2004-02-20	Male	9876543220	rahul.menon@gmail.com	1
	4	Rahul Das	2003-11-30	Male	9876543213	rahul@gmail.com	2
	10	Naveen Kumar	2003-10-11	Male	9876543219	naveen@gmail.com	5
	3	Meera Nair	2004-01-15	Female	9876543212	meera@gmail.com	2
	7	Keerthana S	2004-09-25	Female	9876543216	keerthana@gmail.com	1

Student 2 x

Output

LIKE Operation

```

257 • SELECT *
258 FROM Student
259 WHERE student_name LIKE 'A%';

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	student_id	student_name	date_of_birth	gender	phone	email	department_id
▶	1	Ananya Raj	2004-05-12	Female	9998887770	ananya@gmail.com	1
	2	Arjun Kumar	2003-08-21	Male	9876543211	arjun@gmail.com	1
	8	Adithya P	2003-12-05	Male	9876543217	adithya@gmail.com	4
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

BETWEEN Operation

```

260 • SELECT *
261 FROM Student
262 WHERE date_of_birth BETWEEN '2004-01-01' AND '2004-12-31';

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	student_id	student_name	date_of_birth	gender	phone	email	department_id
▶	1	Ananya Raj	2004-05-12	Female	9998887770	ananya@gmail.com	1
	3	Meera Nair	2004-01-15	Female	9876543212	meera@gmail.com	2
	5	Sneha Thomas	2004-07-09	Female	9876543214	sneha@gmail.com	3
	7	Keerthana S	2004-09-25	Female	9876543216	keerthana@gmail.com	1
	9	Fathima Ali	2004-06-14	Female	9876543218	fathima@gmail.com	1
	13	Rahul Menon	2004-02-20	Male	9876543220	rahul.menon@gmail.com	1
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Student 5 x

Screenshot 7 – Aggregate Functions

COUNT()

```
263 • SELECT COUNT(*) AS total_records
264 FROM Marks;
265
```

Result Grid	Filter Rows:	Export:
total_records		
▶ 17		

t

SUM()

```
263 • SELECT SUM(total_marks) AS total_marks
264 FROM Marks;
```

Result Grid	Filter Rows:	Export:
total_marks		
▶ 1501.00		

it

AVG()

```
264 FROM Marks;
265 • SELECT AVG(total_marks) AS average_marks
266 FROM Marks;
267
```

Result Grid	Filter Rows:	Export:	Wrap C
average_marks			
▶ 88.294118			

ent

MAX()

```
265 • SELECT MAX(total_marks) AS highest_marks
266 FROM Marks;
```

Result Grid

highest_marks
98.00

t

MIN()

```
266 FROM Marks;
267 • SELECT MIN(total_marks) AS lowest_marks
268 FROM Marks;
269
```

Result Grid

lowest_marks
77.00

it

Screenshot 8 – GROUP BY Operation

267 • `SELECT department_id, COUNT(*) AS total_students`
 268 `FROM Student`
 269 `GROUP BY department_id;`
 270

Result Grid | Filter Rows: | Export: | Wrap Cell Conte

	department_id	total_students
▶	1	5
	2	2
	3	2
	4	1
	5	1

it

Screenshot 9 – INNER JOIN Operation

anagei

270 • **SELECT**

271 Student.student_name,

272 Course.course_name,

273 Enrollment.semester,

274 Enrollment.grade

275 **FROM** Student

276 **INNER JOIN** Enrollment

277 **ON** Student.student_name = Enrollment.student_name

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	student_name	course_name	semester	grade
▶	Ananya Raj	Database Management Systems	5	A
	Ananya Raj	Web Technology	5	A
	Ananya Raj	Python Programming	5	B+
	Arjun Kumar	Database Management Systems	5	B+
	Arjun Kumar	Computer Networks	5	A
	Meera Nair	Data Structures	5	A
	Meera Nair	Java Programming	5	A

Result 12 ×

Output

ment

```

276 INNER JOIN Enrollment
277     ON Student.student_id = Enrollment.student_id
278 INNER JOIN Course
279     ON Enrollment.course_id = Course.course_id;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

student_name	course_name	semester	grade
Meera Nair	Java Programming	5	A
Rahul Das	Data Structures	5	B
Rahul Das	Java Programming	5	B+
Sneha Thomas	Information Security	5	A
Vishnu Menon	Information Security	5	B+
Keerthana S	Database Management Systems	5	A
Keerthana S	Operating Systems	5	A

Result 12 x

Output

Action Output

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

student_name	course_name	semester	grade
Keerthana S	Database Management Systems	5	A
Keerthana S	Operating Systems	5	A
Adithya P	Financial Accounting	5	B+
Fathima Ali	Database Management Systems	5	A+
Fathima Ali	Python Programming	5	A
Naveen Kumar	Discrete Mathematics	5	A

Result 12 x

Output

Screenshot 10 – Final Database / Query Results

```

280 • SELECT 'Department' AS table_name, COUNT(*) AS total_records
281 FROM Department
282
283 UNION ALL
284
285 SELECT 'Student', COUNT(*)
286 FROM Student
287
288 UNION ALL
289
290 SELECT 'Course', COUNT(*)
291 FROM Course
292
293 UNION ALL
294
295 SELECT 'Enrollment', COUNT(*)
296 FROM Enrollment
297
298 UNION ALL
299
300 SELECT 'Marks', COUNT(*)
301 FROM Marks
302
303 UNION ALL
304
305 SELECT 'Faculty', COUNT(*)
306 FROM Faculty
307
308 UNION ALL
309
310 SELECT 'Examination', COUNT(*)
311 FROM Examination
312
313 UNION ALL
314
315 SELECT 'Department', COUNT(*)
316 FROM Department
317
318 UNION ALL
319
320 SELECT 'Student', COUNT(*)
321 FROM Student
322
323 UNION ALL
324
325 SELECT 'Course', COUNT(*)
326 FROM Course
327
328 UNION ALL
329
330 SELECT 'Enrollment', COUNT(*)
331 FROM Enrollment
332
333 UNION ALL
334
335 SELECT 'Marks', COUNT(*)
336 FROM Marks
337
338 UNION ALL
339
340 SELECT 'Faculty', COUNT(*)
341 FROM Faculty
342
343 UNION ALL
344
345 SELECT 'Examination', COUNT(*)
346 FROM Examination
347
348 UNION ALL
349
350 SELECT 'Department', COUNT(*)
351 FROM Department
352
353 UNION ALL
354
355 SELECT 'Student', COUNT(*)
356 FROM Student
357
358 UNION ALL
359
360 SELECT 'Course', COUNT(*)
361 FROM Course
362
363 UNION ALL
364
365 SELECT 'Enrollment', COUNT(*)
366 FROM Enrollment
367
368 UNION ALL
369
370 SELECT 'Marks', COUNT(*)
371 FROM Marks
372
373 UNION ALL
374
375 SELECT 'Faculty', COUNT(*)
376 FROM Faculty
377
378 UNION ALL
379
380 SELECT 'Examination', COUNT(*)
381 FROM Examination
382
383 UNION ALL
384
385 SELECT 'Department', COUNT(*)
386 FROM Department
387
388 UNION ALL
389
390 SELECT 'Student', COUNT(*)
391 FROM Student
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393 UNION ALL
394
395 SELECT 'Course', COUNT(*)
396 FROM Course
397
398 UNION ALL
399
400 SELECT 'Enrollment', COUNT(*)
401 FROM Enrollment
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403 UNION ALL
404
405 SELECT 'Marks', COUNT(*)
406 FROM Marks
407
408 UNION ALL
409
410 SELECT 'Faculty', COUNT(*)
411 FROM Faculty
412
413 UNION ALL
414
415 SELECT 'Examination', COUNT(*)
416 FROM Examination
417
418 UNION ALL
419
420 SELECT 'Department', COUNT(*)
421 FROM Department
422
423 UNION ALL
424
425 SELECT 'Student', COUNT(*)
426 FROM Student
427
428 UNION ALL
429
430 SELECT 'Course', COUNT(*)
431 FROM Course
432
433 UNION ALL
434
435 SELECT 'Enrollment', COUNT(*)
436 FROM Enrollment
437
438 UNION ALL
439
440 SELECT 'Marks', COUNT(*)
441 FROM Marks
442
443 UNION ALL
444
445 SELECT 'Faculty', COUNT(*)
446 FROM Faculty
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448 UNION ALL
449
450 SELECT 'Examination', COUNT(*)
451 FROM Examination
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453 UNION ALL
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455 SELECT 'Department', COUNT(*)
456 FROM Department
457
458 UNION ALL
459
460 SELECT 'Student', COUNT(*)
461 FROM Student
462
463 UNION ALL
464
465 SELECT 'Course', COUNT(*)
466 FROM Course
467
468 UNION ALL
469
470 SELECT 'Enrollment', COUNT(*)
471 FROM Enrollment
472
473 UNION ALL
474
475 SELECT 'Marks', COUNT(*)
476 FROM Marks
477
478 UNION ALL
479
480 SELECT 'Faculty', COUNT(*)
481 FROM Faculty
482
483 UNION ALL
484
485 SELECT 'Examination', COUNT(*)
486 FROM Examination
487
488 UNION ALL
489
490 SELECT 'Department', COUNT(*)
491 FROM Department
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493 UNION ALL
494
495 SELECT 'Student', COUNT(*)
496 FROM Student
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498 UNION ALL
499
500 SELECT 'Course', COUNT(*)
501 FROM Course
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503 UNION ALL
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505 SELECT 'Enrollment', COUNT(*)
506 FROM Enrollment
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508 UNION ALL
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510 SELECT 'Marks', COUNT(*)
511 FROM Marks
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513 UNION ALL
514
515 SELECT 'Faculty', COUNT(*)
516 FROM Faculty
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518 UNION ALL
519
520 SELECT 'Examination', COUNT(*)
521 FROM Examination
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523 UNION ALL
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525 SELECT 'Department', COUNT(*)
526 FROM Department
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528 UNION ALL
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530 SELECT 'Student', COUNT(*)
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533 UNION ALL
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535 SELECT 'Course', COUNT(*)
536 FROM Course
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538 UNION ALL
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540 SELECT 'Enrollment', COUNT(*)
541 FROM Enrollment
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543 UNION ALL
544
545 SELECT 'Marks', COUNT(*)
546 FROM Marks
547
548 UNION ALL
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550 SELECT 'Faculty', COUNT(*)
551 FROM Faculty
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553 UNION ALL
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555 SELECT 'Examination', COUNT(*)
556 FROM Examination
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558 UNION ALL
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560 SELECT 'Department', COUNT(*)
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563 UNION ALL
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565 SELECT 'Student', COUNT(*)
566 FROM Student
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568 UNION ALL
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570 SELECT 'Course', COUNT(*)
571 FROM Course
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573 UNION ALL
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575 SELECT 'Enrollment', COUNT(*)
576 FROM Enrollment
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578 UNION ALL
579
580 SELECT 'Marks', COUNT(*)
581 FROM Marks
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583 UNION ALL
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585 SELECT 'Faculty', COUNT(*)
586 FROM Faculty
587
588 UNION ALL
589
590 SELECT 'Examination', COUNT(*)
591 FROM Examination
592
593 UNION ALL
594
595 SELECT 'Department', COUNT(*)
596 FROM Department
597
598 UNION ALL
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600 SELECT 'Student', COUNT(*)
601 FROM Student
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603 UNION ALL
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605 SELECT 'Course', COUNT(*)
606 FROM Course
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608 UNION ALL
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610 SELECT 'Enrollment', COUNT(*)
611 FROM Enrollment
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613 UNION ALL
614
615 SELECT 'Marks', COUNT(*)
616 FROM Marks
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618 UNION ALL
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620 SELECT 'Faculty', COUNT(*)
621 FROM Faculty
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623 UNION ALL
624
625 SELECT 'Examination', COUNT(*)
626 FROM Examination
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628 UNION ALL
629
630 SELECT 'Department', COUNT(*)
631 FROM Department
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633 UNION ALL
634
635 SELECT 'Student', COUNT(*)
636 FROM Student
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638 UNION ALL
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640 SELECT 'Course', COUNT(*)
641 FROM Course
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643 UNION ALL
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645 SELECT 'Enrollment', COUNT(*)
646 FROM Enrollment
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648 UNION ALL
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650 SELECT 'Marks', COUNT(*)
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653 UNION ALL
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655 SELECT 'Faculty', COUNT(*)
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658 UNION ALL
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660 SELECT 'Examination', COUNT(*)
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663 UNION ALL
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665 SELECT 'Department', COUNT(*)
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668 UNION ALL
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670 SELECT 'Student', COUNT(*)
671 FROM Student
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673 UNION ALL
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675 SELECT 'Course', COUNT(*)
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678 UNION ALL
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680 SELECT 'Enrollment', COUNT(*)
681 FROM Enrollment
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683 UNION ALL
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685 SELECT 'Marks', COUNT(*)
686 FROM Marks
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688 UNION ALL
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690 SELECT 'Faculty', COUNT(*)
691 FROM Faculty
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693 UNION ALL
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695 SELECT 'Examination', COUNT(*)
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698 UNION ALL
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700 SELECT 'Department', COUNT(*)
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703 UNION ALL
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705 SELECT 'Student', COUNT(*)
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710 SELECT 'Course', COUNT(*)
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713 UNION ALL
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715 SELECT 'Enrollment', COUNT(*)
716 FROM Enrollment
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718 UNION ALL
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720 SELECT 'Marks', COUNT(*)
721 FROM Marks
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723 UNION ALL
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725 SELECT 'Faculty', COUNT(*)
726 FROM Faculty
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728 UNION ALL
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730 SELECT 'Examination', COUNT(*)
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733 UNION ALL
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735 SELECT 'Department', COUNT(*)
736 FROM Department
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738 UNION ALL
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740 SELECT 'Student', COUNT(*)
741 FROM Student
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743 UNION ALL
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745 SELECT 'Course', COUNT(*)
746 FROM Course
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748 UNION ALL
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750 SELECT 'Enrollment', COUNT(*)
751 FROM Enrollment
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753 UNION ALL
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755 SELECT 'Marks', COUNT(*)
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758 UNION ALL
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760 SELECT 'Faculty', COUNT(*)
761 FROM Faculty
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763 UNION ALL
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765 SELECT 'Examination', COUNT(*)
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768 UNION ALL
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770 SELECT 'Department', COUNT(*)
771 FROM Department
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773 UNION ALL
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775 SELECT 'Student', COUNT(*)
776 FROM Student
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778 UNION ALL
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780 SELECT 'Course', COUNT(*)
781 FROM Course
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783 UNION ALL
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785 SELECT 'Enrollment', COUNT(*)
786 FROM Enrollment
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788 UNION ALL
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790 SELECT 'Marks', COUNT(*)
791 FROM Marks
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793 UNION ALL
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795 SELECT 'Faculty', COUNT(*)
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798 UNION ALL
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800 SELECT 'Examination', COUNT(*)
801 FROM Examination
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803 UNION ALL
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805 SELECT 'Department', COUNT(*)
806 FROM Department
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808 UNION ALL
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810 SELECT 'Student', COUNT(*)
811 FROM Student
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813 UNION ALL
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815 SELECT 'Course', COUNT(*)
816 FROM Course
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818 UNION ALL
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820 SELECT 'Enrollment', COUNT(*)
821 FROM Enrollment
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823 UNION ALL
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825 SELECT 'Marks', COUNT(*)
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828 UNION ALL
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830 SELECT 'Faculty', COUNT(*)
831 FROM Faculty
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833 UNION ALL
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835 SELECT 'Examination', COUNT(*)
836 FROM Examination
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838 UNION ALL
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840 SELECT 'Department', COUNT(*)
841 FROM Department
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843 UNION ALL
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845 SELECT 'Student', COUNT(*)
846 FROM Student
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848 UNION ALL
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850 SELECT 'Course', COUNT(*)
851 FROM Course
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853 UNION ALL
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855 SELECT 'Enrollment', COUNT(*)
856 FROM Enrollment
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858 UNION ALL
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860 SELECT 'Marks', COUNT(*)
861 FROM Marks
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863 UNION ALL
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865 SELECT 'Faculty', COUNT(*)
866 FROM Faculty
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868 UNION ALL
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870 SELECT 'Examination', COUNT(*)
871 FROM Examination
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873 UNION ALL
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875 SELECT 'Department', COUNT(*)
876 FROM Department
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878 UNION ALL
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880 SELECT 'Student', COUNT(*)
881 FROM Student
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883 UNION ALL
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885 SELECT 'Course', COUNT(*)
886 FROM Course
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888 UNION ALL
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890 SELECT 'Enrollment', COUNT(*)
891 FROM Enrollment
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893 UNION ALL
894
895 SELECT 'Marks', COUNT(*)
896 FROM Marks
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898 UNION ALL
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900 SELECT 'Faculty', COUNT(*)
901 FROM Faculty
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903 UNION ALL
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905 SELECT 'Examination', COUNT(*)
906 FROM Examination
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908 UNION ALL
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910 SELECT 'Department', COUNT(*)
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913 UNION ALL
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915 SELECT 'Student', COUNT(*)
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925 SELECT 'Enrollment', COUNT(*)
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943 UNION ALL
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955 SELECT 'Course', COUNT(*)
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961 FROM Enrollment
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963 UNION ALL
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965 SELECT 'Marks', COUNT(*)
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968 UNION ALL
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980 SELECT 'Department', COUNT(*)
981 FROM Department
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983 UNION ALL
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985 SELECT 'Student', COUNT(*)
986 FROM Student
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988 UNION ALL
989
990 SELECT 'Course', COUNT(*)
991 FROM Course
992
993 UNION ALL
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995 SELECT 'Enrollment', COUNT(*)
996 FROM Enrollment
997
998 UNION ALL
999
1000 SELECT 'Marks', COUNT(*)
1001 FROM Marks
1002
1003 UNION ALL
1004
1005 SELECT 'Faculty', COUNT(*)
1006 FROM Faculty
1007
1008 UNION ALL
1009
1010 SELECT 'Examination', COUNT(*)
1011 FROM Examination
1012
1013 UNION ALL
1014
1015 SELECT 'Department', COUNT(*)
1016 FROM Department
1017
1018 UNION ALL
1019
1020 SELECT 'Student', COUNT(*)
1021 FROM Student
1022
1023 UNION ALL
1024
1025 SELECT 'Course', COUNT(*)
1026 FROM Course
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1028 UNION ALL
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1030 SELECT 'Enrollment', COUNT(*)
1031 FROM Enrollment
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1033 UNION ALL
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1038 UNION ALL
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1040 SELECT 'Faculty', COUNT(*)
1041 FROM Faculty
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1043 UNION ALL
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1045 SELECT 'Examination', COUNT(*)
1046 FROM Examination
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1048 UNION ALL
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1050 SELECT 'Department', COUNT(*)
1051 FROM Department
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1053 UNION ALL
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1055 SELECT 'Student', COUNT(*)
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1060 SELECT 'Course', COUNT(*)
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1063 UNION ALL
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1065 SELECT 'Enrollment', COUNT(*)
1066 FROM Enrollment
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1068 UNION ALL
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1070 SELECT 'Marks', COUNT(*)
1071 FROM Marks
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1073 UNION ALL
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1075 SELECT 'Faculty', COUNT(*)
1076 FROM Faculty
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1078 UNION ALL
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1080 SELECT 'Examination', COUNT(*)
1081 FROM Examination
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1083 UNION ALL
1084
1085 SELECT 'Department', COUNT(*)
1086 FROM Department
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1088 UNION ALL
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1090 SELECT 'Student', COUNT(*)
1091 FROM Student
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1093 UNION ALL
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1095 SELECT 'Course', COUNT(*)
1096 FROM Course
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1098 UNION ALL
1099
1100 SELECT 'Enrollment', COUNT(*)
1101 FROM Enrollment
1102
1103 UNION ALL
1104
1105 SELECT 'Marks', COUNT(*)
1106 FROM Marks
1107
1108 UNION ALL
1109
1110 SELECT 'Faculty', COUNT(*)
1111 FROM Faculty
1112
1113 UNION ALL
1114
1115 SELECT 'Examination', COUNT(*)
1116 FROM Examination
1117
1118 UNION ALL
1119
1120 SELECT 'Department', COUNT(*)
1121 FROM Department
1122
1123 UNION ALL
1124
1125 SELECT 'Student', COUNT(*)
1126 FROM Student
1127
1128 UNION ALL
1129
1130 SELECT 'Course', COUNT(*)
1131 FROM Course
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1133 UNION ALL
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1135 SELECT 'Enrollment', COUNT(*)
1136 FROM Enrollment
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1138 UNION ALL
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1140 SELECT 'Marks', COUNT(*)
1141 FROM Marks
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1143 UNION ALL
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1145 SELECT 'Faculty', COUNT(*)
1146 FROM Faculty
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1148 UNION ALL
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1150 SELECT 'Examination', COUNT(*)
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1153 UNION ALL
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1155 SELECT 'Department', COUNT(*)
1156 FROM Department
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1158 UNION ALL
1159
1160 SELECT 'Student', COUNT(*)
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1163 UNION ALL
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1165 SELECT 'Course', COUNT(*)
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1168 UNION ALL
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1171 FROM Enrollment
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1173 UNION ALL
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1175 SELECT 'Marks', COUNT(*)
1176 FROM Marks
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1178 UNION ALL
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1180 SELECT 'Faculty', COUNT(*)
1181 FROM Faculty
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1183 UNION ALL
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1185 SELECT 'Examination', COUNT(*)
1186 FROM Examination
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1188 UNION ALL
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1190 SELECT 'Department', COUNT(*)
1191 FROM Department
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1193 UNION ALL
1194
1195 SELECT 'Student', COUNT(*)
1196 FROM Student
1197
1198 UNION ALL
1199
1200 SELECT 'Course', COUNT(*)
1201 FROM Course
1202
1203 UNION ALL
1204
1205 SELECT 'Enrollment', COUNT(*)
1206 FROM Enrollment
1207
1208 UNION ALL
1209
1210 SELECT 'Marks', COUNT(*)
1211 FROM Marks
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1213 UNION ALL
1214
1215 SELECT 'Faculty', COUNT(*)
1216 FROM Faculty
1217
1218 UNION ALL
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1220 SELECT 'Examination', COUNT(*)
1221 FROM Examination
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1223 UNION ALL
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1225 SELECT 'Department', COUNT(*)
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1228 UNION ALL
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1230 SELECT 'Student', COUNT(*)
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1233 UNION ALL
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1235 SELECT 'Course', COUNT(*)
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1238 UNION ALL
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1240 SELECT 'Enrollment', COUNT(*)
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1243 UNION ALL
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1245 SELECT 'Marks', COUNT(*)
1246 FROM Marks
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1248 UNION ALL
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1250 SELECT 'Faculty', COUNT(*)
1251 FROM Faculty
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1253 UNION ALL
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1255 SELECT 'Examination', COUNT(*)
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1258 UNION ALL
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1260 SELECT 'Department', COUNT(*)
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1263 UNION ALL
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1265 SELECT 'Student', COUNT(*)
1266 FROM Student
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1268 UNION ALL
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1270 SELECT 'Course', COUNT(*)
1271 FROM Course
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1273 UNION ALL
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1275 SELECT 'Enrollment', COUNT(*)
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1278 UNION ALL
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1280 SELECT 'Marks', COUNT(*)
1281 FROM Marks
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1283 UNION ALL
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1285 SELECT 'Faculty', COUNT(*)
1286 FROM Faculty
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1288 UNION ALL
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1290 SELECT 'Examination', COUNT(*)
1291 FROM Examination
1292
1293 UNION ALL
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1295 SELECT 'Department', COUNT(*)
1296 FROM Department
1297
1298 UNION ALL
1299
1300 SELECT 'Student', COUNT(*)
1301 FROM Student
1302
1303 UNION ALL
1304
1305 SELECT 'Course', COUNT(*)
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293 UNION ALL
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295 SELECT 'Enrollment', COUNT(*)
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300 SELECT 'Marks', COUNT(*)

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```

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300 SELECT 'Marks', COUNT(*)
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303 UNION ALL
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305 SELECT 'Faculty', COUNT(*)
306 FROM Faculty

```

table_name	total_records
Enrollment	1000
Marks	1000
Faculty	1000

```

305 SELECT 'Faculty', COUNT(*)
306 FROM Faculty
307
308 UNION ALL
309
310 SELECT 'Examination', COUNT(*)
311 FROM Examination;

```

5. Technical Explanation

Database Design

The **Student Database Management System** is designed using a relational database model in MySQL. The database contains **seven interconnected tables** that organize student, academic, faculty, and examination information.

Tables in the Database

1. **Department** – Stores department details and HOD information.
2. **Student** – Stores student personal and contact details.
3. **Course** – Stores course names, codes, credits, and department details.
4. **Enrollment** – Maintains student course enrollment and grades.
5. **Marks** – Stores internal, external, and total marks.
6. **Faculty** – Stores faculty details and their departments.
7. **Examination** – Stores examination names, types, dates, and related courses.

Relationships

- Department → Student
- Department → Course
- Department → Faculty
- Student → Enrollment
- Course → Enrollment
- Enrollment → Marks
- Course → Examination

Primary keys uniquely identify records, while **foreign keys** establish relationships between the tables and maintain data integrity.

Keys and Constraints



The Student Database Management System uses keys and constraints to maintain **data accuracy, uniqueness, and relationships** between tables.

Keys Used

- **Primary Key (PK):** Uniquely identifies each record in a table.
- **Foreign Key (FK):** Connects related tables and maintains referential integrity.

Constraints Used

- **NOT NULL:** Ensures that important fields cannot be left empty.
- **UNIQUE:** Ensures that values such as course_code are not duplicated.
- **AUTO_INCREMENT:** Automatically generates unique ID values.
- **PRIMARY KEY:** Ensures each record has a unique identifier.
- **FOREIGN KEY:** Ensures that referenced records exist in the related table.

Primary and Foreign Keys

Table	Primary Key	Foreign Key
Department	department_id	—
Student	student_id	department_id
Course	course_id	department_id
Enrollment	enrollment_id	student_id, course_id
Marks	mark_id	enrollment_id
Faculty	faculty_id	department_id
Examination	exam_id	course_id

Conclusion: These keys and constraints help maintain **data integrity, consistency, and reliable relationships** within the database.



SQL Operations

The basic **CRUD (Create, Read, Update, Delete)** operations were implemented in the Student Database Management System using SQL.

1. INSERT Operation

The INSERT statement was used to add new records to the database.

```
INSERT INTO Student
(student_name, date_of_birth, gender, phone, email,
department_id)
VALUES
('Rahul Menon', '2004-02-20', 'Male', '9876543220',
'rahul.menon@gmail.com', 1);
```

Purpose: Adds a new student record to the Student table.

2. SELECT Operation

The SELECT statement was used to retrieve and display records.

```
SELECT * FROM Student;
```

Purpose: Displays all records and columns from the Student table.

3. UPDATE Operation

The UPDATE statement was used to modify existing information.

```
UPDATE Student
SET phone = '9998887770'
WHERE student_id = 1;
```



Purpose: Updates the phone number of the student whose ID is 1.

4. DELETE Operation

The DELETE statement was used to remove an unwanted record.

```
DELETE FROM Student  
WHERE student_id = 12;
```

Purpose: Deletes the student record with student_id = 12.

Summary

Operation	SQL Command	Purpose
INSERT	INSERT INTO	Adds new records
SELECT	SELECT	Retrieves records
UPDATE	UPDATE	Modifies existing records
DELETE	DELETE	Removes records

These four operations demonstrate the basic **CRUD functionality** of the database system.

.

Data Retrieval

These SQL clauses were used to **filter, sort, search, and organize data** in the Student Database Management System.

1. WHERE

WHERE is used to retrieve records that satisfy a specific condition.



```
SELECT * FROM Student  
WHERE department_id = 1;
```

Purpose: Displays students belonging to department 1.

2. ORDER BY

ORDER BY is used to sort records in ascending or descending order.

```
SELECT * FROM Student  
ORDER BY student_name DESC;
```

Purpose: Displays student names in descending alphabetical order.

3. LIKE

LIKE is used to search for records matching a specific pattern.

```
SELECT * FROM Student  
WHERE student_name LIKE 'A%';
```

Purpose: Displays students whose names start with the letter **A**.

4. BETWEEN

BETWEEN is used to retrieve values within a specified range.

```
SELECT * FROM Student  
WHERE date_of_birth BETWEEN '2004-01-01' AND '2004-12-31';
```

Purpose: Displays students born during the year **2004**.

5. GROUP BY

GROUP BY is used to group records based on a particular column, often with aggregate functions.

```
SELECT department_id, COUNT(*) AS total_students  
FROM Student  
GROUP BY department_id;
```

Purpose: Displays the number of students in each department.

Summary

Clause	Purpose
WHERE	Filters records
ORDER BY	Sorts records
LIKE	Searches using a pattern
BETWEEN	Filters within a range
GROUP BY	Groups records for analysis

Aggregate Functions and Joins

1. Aggregate Functions

Aggregate functions are used to **perform calculations on a set of records and return a single result**. In this project, the following aggregate functions were used:

- COUNT() – Counts the number of records.
- SUM() – Calculates the total of values.
- AVG() – Calculates the average value.
- MAX() – Finds the highest value.
- MIN() – Finds the lowest value.

Example:

```
SELECT COUNT(*) AS total_records FROM Marks;  
SELECT SUM(total_marks) AS total_marks FROM Marks;
```



```
SELECT AVG(total_marks) AS average_marks FROM Marks;  
SELECT MAX(total_marks) AS highest_marks FROM Marks;  
SELECT MIN(total_marks) AS lowest_marks FROM Marks;
```

Purpose: These functions help analyze and summarize academic marks stored in the database.

2. INNER JOIN

INNER JOIN is used to **retrieve related information from two or more tables** by matching their related key columns.

Example using three tables:

```
SELECT  
    Student.student_name,  
    Course.course_name,  
    Enrollment.semester,  
    Enrollment.grade  
FROM Student  
INNER JOIN Enrollment  
    ON Student.student_id = Enrollment.student_id  
INNER JOIN Course  
    ON Enrollment.course_id = Course.course_id;
```

Purpose: This query combines the Student, Enrollment, and Course tables to display the **student name, enrolled course, semester, and grade** together.

Another example using four tables:

```
SELECT  
    Student.student_name,  
    Course.course_name,  
    Marks.internal_marks,  
    Marks.external_marks,  
    Marks.total_marks
```



```
FROM Student
INNER JOIN Enrollment
    ON Student.student_id = Enrollment.student_id
INNER JOIN Course
    ON Enrollment.course_id = Course.course_id
INNER JOIN Marks
    ON Enrollment.enrollment_id = Marks.enrollment_id;
```

Purpose: This retrieves a student's **name, course, internal marks, external marks, and total marks** from multiple related tables.

Summary

Aggregate functions are used to summarize and analyze numerical data, while **INNER JOIN** is used to combine related records from multiple tables using their primary and foreign key relationships.

6.Conclusion

Conclusion

The **Student Database Management System** was successfully developed using **MySQL** to manage student and academic information in a structured relational database. The database consists of seven interconnected tables: **Department, Student, Course, Enrollment, Marks, Faculty, and Examination**.

The major features implemented include student record management, department and course management, enrollment tracking, marks management, faculty information, and examination details. The project also demonstrates important SQL operations such as **CREATE, INSERT, SELECT, UPDATE, DELETE, ALTER TABLE, WHERE, ORDER BY, LIKE, BETWEEN, GROUP BY, aggregate functions, INNER JOIN, and UNION ALL**.



Through this project, the concepts of **relational database design, primary and foreign keys, constraints, table relationships, data integrity, CRUD operations, data filtering, sorting, grouping, aggregation, and joining multiple tables** were learned and practically implemented using MySQL Workbench.

Future Enhancements

The Student Database Management System can be further improved by adding the following features:

- **Student Login and Authentication** – Provide secure login facilities for students, faculty, and administrators.
- **Online Course Enrollment** – Allow students to enroll in courses through an online interface.
- **Automated Examination Notifications** – Send reminders about upcoming examinations and important academic dates.
- **Academic Performance Reports** – Generate detailed reports of students' marks, grades, and overall performance.
- **Online Grade Management** – Allow authorized faculty members to enter and update student marks securely.
- **Faculty Management** – Add features for managing faculty assignments, courses, and department responsibilities.
- **Web or Mobile Application Integration** – Connect the database with a web or mobile application for easy access.
- **Automated Database Backup and Recovery** – Implement regular backups and recovery mechanisms to protect database information.
- **Dashboard and Analytics** – Provide graphical dashboards for viewing student, course, enrollment, and examination statistics.

These improvements can make the system more **secure, accessible, automated, and user-friendly** in the future.